

Problem Set 4

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Problem 1: Edit Distance. Implement the edit distance algorithm. (*LeetCode 72 – Edit Distance*)

Problem 2: Minimum Weighted Sum of Squares. Given an array A of n numbers, you must **rearrange** (permute) its entries into positions $1, 2, \dots, n$ to **minimize**

$$\sum_{i=1}^n i \cdot A[i]^2$$

where $A[i]$ denotes the entry placed in position i after rearranging.

Example: $A = [3, 1, 2]$. Placing $[1, 2, 3]$ in positions $1, 2, 3$ gives $1 \cdot 1^2 + 2 \cdot 2^2 + 3 \cdot 3^2 = 1 + 8 + 27 = 36$, while placing $[3, 2, 1]$ gives $1 \cdot 9 + 2 \cdot 4 + 3 \cdot 1 = 9 + 8 + 3 = 20$. Can you always find the best ordering without trying all $n!$ permutations? Prove the correctness. Hint: Prove that for all i and $i + 1$, $A[i]^2 \geq A[i + 1]^2$ in the optimal solution (if not, what happens to the sum if you swap them).